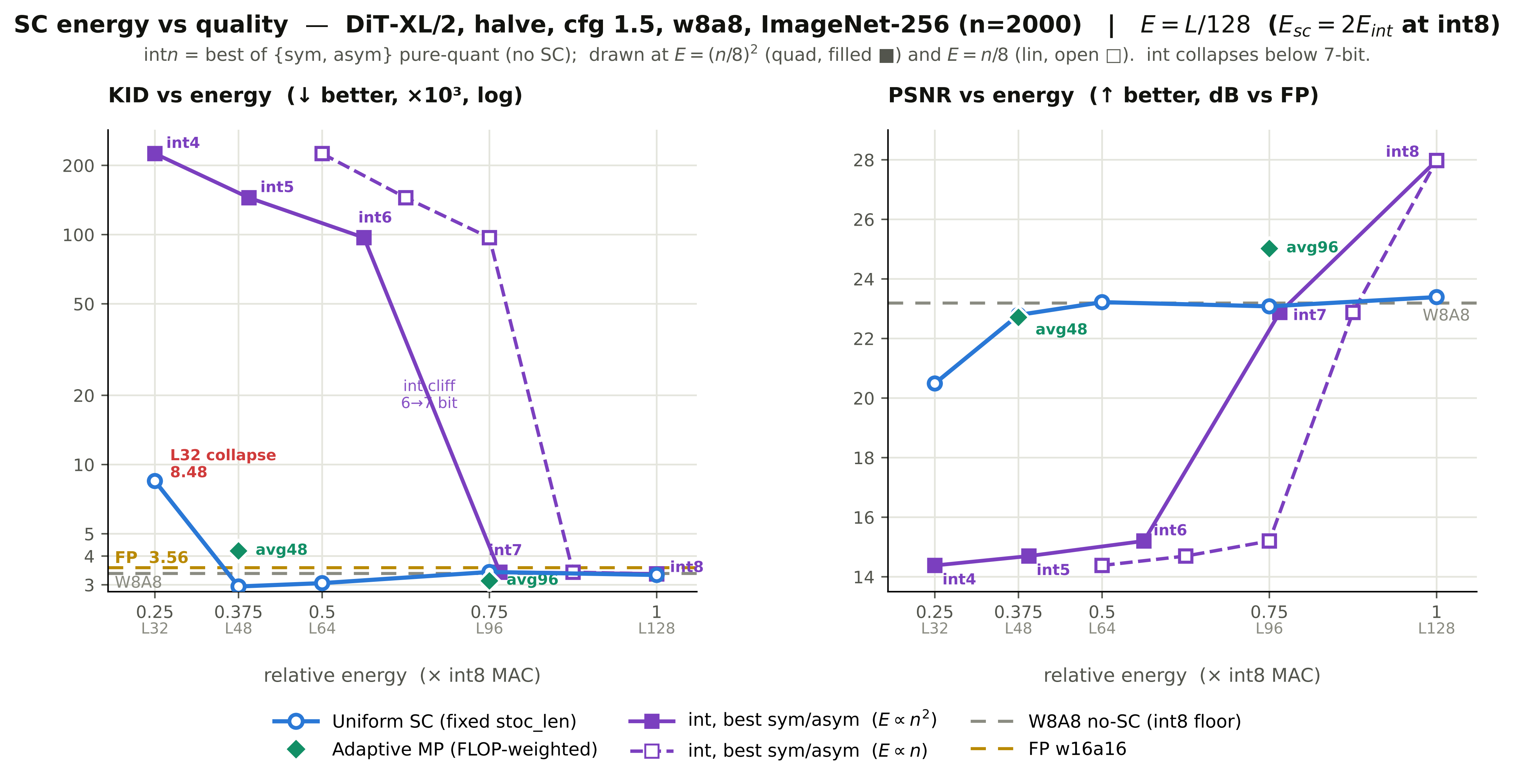

SC energy vs quality — DiT-XL/2, halve, cfg 1.5, w8a8, ImageNet-256 (n=2000) | $E = L/128$ ($E_{sc} = 2E_{int}$ at int8)
int$n$ = best of {sym, asym} pure-quant (no SC); drawn at $E = (n/8)^2$ (quad, filled ■) and $E = n/8$ (lin, open □). int collapses below 7-bit.
KID vs energy (↓ better, ×10³, log)
PSNR vs energy (↑ better, dB vs FP)
int4
int5
int6
int7
int8
int cliff
6→7 bit
L32 collapse
8.48
avg48
avg96
FP 3.56
W8A8
relative energy (× int8 MAC)
0.25
L32
0.375
L48
0.5
L64
0.75
L96
1
L128
Uniform SC (fixed stoc_len)
Adaptive MP (FLOP-weighted)
int, best sym/asym ($E \propto n^2$)
int, best sym/asym ($E \propto n$)
W8A8 no-SC (int8 floor)
FP w16a16